

IPL DATA ANALYSIS PROJECT

Problem Statement | 2008 – 2024



Student / Analyst	: Shafaana Khaalida
Project Type	: End-to-End Data Analysis
Dataset	: IPL Ball-by-Ball 2008 – 2026 260,000+ records
Tools Used	: Python • SQL • Power BI • DAX
Source	: Kaggle / Cricsheet / GitHub
Submission	: Resume Portfolio + GitHub

PROBLEM STATEMENT

The Indian Premier League (IPL) generates enormous amounts of match data every season. Despite its availability, this data remains largely unexplored in a structured analytical form. This project performs a comprehensive end-to-end data analysis on IPL match and ball-by-ball records from 2008 to 2024, covering over **260,000 deliveries** across **1,095+ matches**. Using Python, SQL, and Power BI, the project uncovers **player performance trends**, **team win patterns**, **toss impact**, **venue behaviour**, and **season-wise evolution** — delivering actionable insights through an interactive dashboard.

OBJECTIVES

- **Clean & preprocess** — raw IPL datasets — handle nulls, duplicates, type mismatches, outliers
- **Model the data** — into a Star Schema (fact + dimension tables) and write SQL JOINS
- **Perform analysis** — answering 10 targeted business questions using Python and DAX
- **Build a dashboard** — in Power BI with 3 interactive report pages and dynamic slicers
- **Publish findings** — on GitHub with a clean README for portfolio/resume use

10 ANALYSIS QUESTIONS

Q1

BATTING

Who are the top 10 run-scorers in IPL history with their batting average and

Q2

BATTING

Which batsmen hit the most sixes and fours — and in which venues do they

Q3

BOWLING

Who are the top 10 wicket-takers and what is their economy rate and bowling

Q4

BOWLING

Which bowlers are the most economical in death overs (overs

Q5

TEAM

Which team has the highest win percentage across all IPL seasons?

Q6

TEAM

Which team performs better while chasing vs defending — and by how

Q7

MATCH

Does winning the toss increase the chances of winning the match by

Q8

VENUE

What is the average first-innings score per venue and how does it affect the

Q9

TRENDS

How has the average team score per season evolved from 2008 to 2024?

Q10

PLAYER

Which players win the most Player of the Match awards — is it always top

TOOLS & DATASETS

Tool / Source	Purpose	Detail
Python (Pandas, NumPy)	Data cleaning & analysis	EDA, descriptive & comparative analysis
Matplotlib / Seaborn	Visualisation	Bar, line, heatmap, scatter plots
SQL (SQLite)	Data modelling & JOINS	Star Schema — fact + 4 dimension tables
Power BI + DAX	Dashboard & KPIs	3-page interactive report with slicers
Kaggle / GitHub	Dataset source	matches.csv + deliveries.csv (2008-2024)
GitHub	Portfolio	README, Jupyter Notebook, .pbix file

EXPECTED OUTCOMES

Deliverable	Description
Cleaned dataset	Fully cleaned matches.csv + deliveries.csv ready for analysis
SQL Star Schema	4 dimension tables + 1 fact table with working JOIN queries
Python Notebook	Jupyter Notebook with EDA, charts, and markdown explanations
Power BI Dashboard	3-page report: Team Overview, Player Leaderboard, Venue & Toss
GitHub Repository	Public repo with README, dataset, notebook, and .pbix file
Resume Bullet	Quantified achievement: 260K+ records, 10 insights, 3-page dashboard